FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2021

TEST DATE: 08-02-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. $W_{man} + Wg = \Delta kE \qquad KE_i = 0 \;\&\; KE_f = 0$

So $W_{man} = mgh$

Where h is height gained by C.M

$= 80 \times 10 \times 0.9 = 720J$

2. C

Sol. $W_f = -\frac{1}{2}mv_0^2$

$W_f \; on \; A + W_f \; on \; B = 0$

So $W_f \; on \; B = \frac{1}{2}mv_0^2$

By work energy theorem on B

$W_{spring} + W_f = -\frac{1}{2} \times 2m \times W_f \; on \; B = \frac{1}{2}mv_0^2$

$W_{spring} = -\frac{3}{2}mv_0^2$

3. B

Sol. $x = \frac{C}{T}$

$$\frac{x_1}{x_2}=\frac{T_2}{T_1}=\frac{(273-173)}{(273-73)}$$
$$x_2=0.0120$$

4. A

Sol. $$\frac{GMm}{r^2}=m\omega^2 r \Rightarrow r^3=\frac{GM}{\omega^2}=\frac{gR^2}{\omega^2}$$
$$\therefore r=\left(gR^2/\omega^2\right)^{1/3}$$

5. C

Sol. Time period of revolving satellite $T=2\pi\sqrt{R^3/GM}$ for satellite revolving close to earth $R_e=R$.

$$\mathrm{T}=2\pi\sqrt{\mathrm{R}^3{}_e/\mathrm{G}4/3\pi\mathrm{R}_e^3\rho}$$

So, $T\alpha 1/\sqrt{\rho}$

As T is equal, ρ is also equal.

6. B

Sol. Transational k. E of a molecule $3/2\mathrm{kT}$

Transational k. E of a mole $3/2\mathrm{RT}$

$$=\frac{3}{2}\mathrm{PV}=\frac{3}{2}\frac{\mathrm{P}_0}{8}\mathrm{V}_0$$

7. D

Sol. Maximum percentage error in $\mathrm{k}=\left(\frac{\Delta a}{a}+\frac{\Delta b}{b}+\frac{3\Delta c}{c}\right)\times 100$

$$=1\%+1\%+3\times 2\%$$
$$=8\%$$

8. D

Sol. Actual frequency will be heard when velocity of motor cyclist is perpendicular to line joining the source and motor cyclist. It will happen 2 times per round.

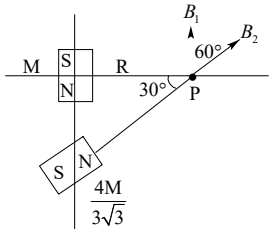
Number of rounds in a hour $=3600/4\pi$ (where 4π is time for one round)

≈ 287

Answer is double of 287 ie 574

9. C

Sol. $$\mathrm{B}_1=\frac{\mu_0}{4\pi}\frac{M}{R^3}=B_0$$
$$\mathrm{B}_2=\frac{2\mu_0}{4\pi}.4M/3\sqrt{3}/\left(2R/\sqrt{3}\right)^3$$
$$=\frac{\mu_0}{4\pi}\frac{M}{R^3}=B_0 \Rightarrow B_{net}=2B_0\cos 60°/2 \Rightarrow B_0\sqrt{3}$$



10. C

Sol. Using, $\Delta E = \frac{hc}{\lambda} = \frac{1242 eV - nm}{0.021 nm}$

$= \frac{1242}{21 \times 10^{-3}} eV = 59 KeV$

11. A

Sol. Using, $\lambda = h / p$

$p = \sqrt{2mK}$, where K is kinetic energy

As kinetic energy is doubled and velocity is also doubled mass becomes half, hence momentum remains unchanged and so is de-Broglie's wavelength.

12. A

13. A

Sol. Given $\sigma = \frac{q_1}{4\pi R^2} = \frac{q_2}{4\pi r^2}$

$V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{R}, V_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r}, \frac{V_2}{V_1} = \frac{R}{r}$

14. A

Sol. As the system is already balanced about Y-axis, the mass has to be kept on the line

OE Using $X_{cm} = \frac{2\times(-2) + 2\times(-2) + 8(2)}{M} = 0$

$\Rightarrow 8x = 8 \Rightarrow x = 1m$

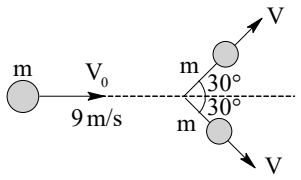
15. D

Sol. Using momentum conservation

$mv_0 = mv\cos 30° + mv\cos 30°$

$9 = \frac{\sqrt{3}}{2} v \times 2 \Rightarrow 9 = \sqrt{3} v$

$\Rightarrow v = 3\sqrt{3}$ m/s . As there was no momentum along Y-axis the speed of each ball has to be same.



16. B

Sol. Given $u+v=100\,cm$

Using lens formula with sign convection $\frac{1}{(100-u)} + \frac{1}{u} = \frac{1}{f}$ and $\frac{1}{(60-u)} + \frac{1}{(u+40)} = \frac{1}{f}$

equation them, $u = 30cm$

Now putting in any of the equations $\frac{1}{70} + \frac{1}{30} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{1}{21cm}$

$p = 5\,m^{-1}$

17. D

Sol. The electric potential at a point, $V=-x^2y-xz^3+4$.

The field $\vec{E}=-\vec{\nabla}V=-\left(\frac{\partial V}{\partial x}\hat{i}+\frac{\partial V}{\partial y}\hat{j}+\frac{\partial V}{\partial z}\hat{k}\right)$

$\therefore \vec{E}=\hat{i}\left(2xy+z^3\right)+\hat{j}x^2+\hat{k}\left(3xz^2\right)$

18. D

Sol. Given: $y_1=4\sin 600\pi t, y_2=5\sin 608\pi t$

$\therefore \omega_1=600\pi \ \ or\ 2\pi\upsilon_1=600\pi\ or\ \upsilon_1=300$

$A_1=4$

And $\omega_2=608\pi\ or\ 2\pi\upsilon_2=608\pi\ or\ \upsilon_2=304$

$A_2=5$

Number of beats heard per second

$=\upsilon_2-\upsilon_1=304-300=4$

$\frac{I_{\max}}{I_{\min}}=\frac{\left(A_1+A_2\right)^2}{\left(A_1-A_2\right)^2}=\frac{\left(4+5\right)^2}{\left(4-5\right)^2}=\frac{81}{1}$

19. C

Sol. The internal resistance of the cell is $r=\left(\frac{l_1}{l_2}-1\right)R$

Here, $l_1=3, l_2=2.85\,m, R=9.5\Omega$

$\therefore r=\left(\frac{3}{2.85}-1\right)\left(9.5\Omega\right)=\frac{0.15}{2.85}\times 9.5\Omega=0.5\Omega$

20. C

Sol. Efficiency $\left(\eta\right)$ of a carnot engine is given by $\eta=1-\frac{T_2}{T_1}$, where T_1 is the temperature of the source and T_2 is the temperature of the sink.

Here, $T_2=500K$

$\therefore 0.5=1-\frac{500}{T_1}\Rightarrow T_1=1000K$.

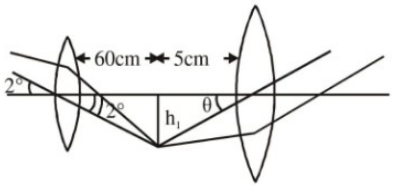
Now, $\eta=0.6=1-\frac{T'_2}{1000}\left(T'_2 \text{ is the new sink temperature}\right)$

$\Rightarrow T'_2=400K$

SECTION – B

21. 00024.00

Sol. $2°=\frac{h_1}{60} \quad .....(1)$



$\theta=\frac{h_1}{5} \quad .....(2)$

Equation (2) ÷ equation (1).....

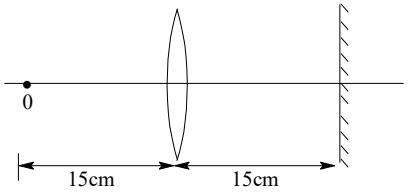
$\Rightarrow \frac{\theta}{2}=\frac{60}{5} \Rightarrow \theta = 24°$

22. 00060.00

Sol. For lens

$\frac{1}{V'}-\frac{1}{(-15)}=\frac{1}{30} \Rightarrow v' = -30cm$

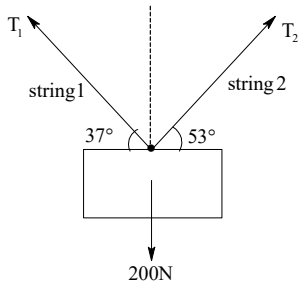
So an image is formed 45cm behind the mirror which will act as an object for lens.



$\frac{1}{V}-\frac{1}{(-60)}=\frac{1}{30} \Rightarrow v = 60cm$

23. 00014.00

Sol. $\frac{4T_1}{5}=\frac{3T_2}{5} \Rightarrow T_1=\frac{3}{4}T_2$



$v=\sqrt{\frac{T}{\mu}} \Rightarrow \frac{v_1}{v_2}=\sqrt{\frac{3}{4}}$

$\ell_1 = 10\times\frac{4}{5}=8m$

$\ell_2 = 10\times\frac{3}{5}=6m \Rightarrow \frac{t_1}{t_2}=\frac{\ell_1/v_1}{\ell_2/v_2}=\frac{8}{6}\sqrt{\frac{4}{3}}=\frac{8}{3\sqrt{3}}$

24. 00001.20

Sol. Using $\mu=\frac{C}{V}$

$\frac{\mu_2}{\mu_1}=\frac{v_1}{v_2} \Rightarrow \mu_2=\left(\frac{v_1}{v_2}\right)\mu_1=\left(\frac{2\times10^8}{2.5\times10^8}\right)\times1.5$

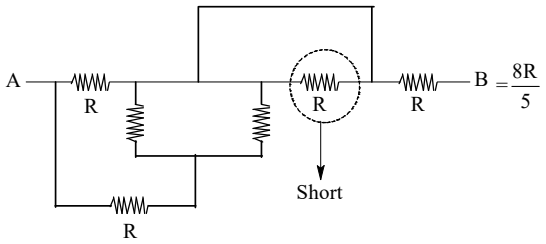
$\mu_2=\frac{6}{5} \Rightarrow \mu_2 = 1.2$

25. 00001.30

26. 00060.00

27. 00001.60

Sol.



28. 00002.00

Sol. $\vec{E}=\dfrac{kq}{\left\{x^2+(x+y)^2\right\}^{3/2}}x\hat{i}+(x+y)\hat{j}+\left[\dfrac{2kp\left(\hat{i}+\hat{j}\right)}{\left(x^2+y^2\right)^{3/2}\sqrt{2}}+\dfrac{kp\left(-\hat{i}+\hat{j}\right)}{\left(x^2+x^2\right)^{3/2}\sqrt{2}}\right]=0$

Take ratio after equating i and j coeff.

$y=2x$

29. 00002.00

Sol. $R=50\Omega$, $X_L=\omega L=50\Omega$

$X_L=\dfrac{1}{\omega C}=100\Omega \quad \Rightarrow Z=\sqrt{R^2+\left(X_L-X_C\right)^2}$

$Z=50\sqrt{2}\Omega$

$I_{rms}=\dfrac{V_{rms}}{Z}=\dfrac{200/\sqrt{2}}{50\sqrt{2}}=2\text{A}$

30. 00400.00

Sol. IIIrd overtone

$\text{f}=\dfrac{4V}{2L}=\dfrac{2V}{L}=\dfrac{2}{1}\sqrt{\dfrac{200}{(5/1000)}}=2(200)$

$f=400\,\text{Hz}$

Chemistry

PART – II

SECTION – A

31. C

32. D

33. A

34. B

35. C

Sol. Guanidine is the most basic compound among the given compounds. Its conjugate acid is most stable due to resonance.

36. D

37. C

38. D

39. A

40. D

41. A

42. D

43. C

44. D

45. C

46. C

47. C

48. B

49. B

50. D

SECTION – B

51. 00031.00

Sol. $C(s)+2H_2(g)\longrightarrow CH_4(g);\quad \Delta H_1=-17\,kcal$ ..... (i)

$2C(s)+3H_2(g)\longrightarrow C_2H_6(g);\quad \Delta H_2=-24\,kcal$ ..... (ii)

$3C(s)+4H_2(g)\longrightarrow C_3H_8(g);\quad \Delta_f H=?$ ..... (iii)

From equations (i) and (ii), we get

Let x kcal mol^{-1} is the energy of $C(s)\longrightarrow C(g)$

Let y kcal mol^{-1} is the BE energy of $(H-H)$

From equation (i), we get

$-17=x+2y-4\times 99$

From equation (ii), we get

$-24=2x+3y-(84+6\times 99)$

Solved for x and y,

$x = 171 kcal, y = 104 kcal$

From equation (iii), we get

$\Delta_f H = 3x + 4y - [2 \times 84 + 8 \times 99]$

$\Rightarrow 31 \text{kcal mol}^{-1}$

52. 00008.93

Sol.

$$N_2O_4 \rightleftharpoons 2NO_3$$

Pressure at equilibrium	0.7	0.3

$$\therefore K_p = \frac{\left(p_{NO_2}\right)^2}{p_{N_2O_4}} = \frac{0.3 \times 0.3}{0.7} = 0.1286 atm$$

Now assume decomposition at 1.0 atm pressure

$$N_2O_4 \rightleftharpoons 2NO_2$$

Initial moles	1	0
Moles at equilibrium	$(1-x)$	$2x$

$$K_p = \frac{\left(n_{NO_2}\right)^2}{n_{N_2O_4}} \times \left(\frac{p}{\sum n}\right)^{\Delta n} = \frac{(2x)^2}{(1-x)} \times \left[\frac{10}{(1+x)}\right]^1$$

$$p'_{NO_2} = \frac{2x}{(1+x)} \times p = \frac{2 \times 0.0565 \times 10}{(1+0.0565)} = 1.07 \text{ atm}$$

$$p'_{N_2O_2} = \left[\frac{1-0.0565}{1+0.0565}\right] \times 10 = 8.93 atm$$

53. 00019.39

Sol. Moles of $H_2 = \frac{W}{\text{Molar weight}} = \frac{20}{2} = 10 \text{ mol}$

Moles of $O_2 = \frac{W}{\text{Molar weight}} = \frac{128}{32} = 4 \text{ mol}$

Total pressure $P = p_{H_2} + p_{O_2}$

$$\left(n_{H_2} + n_{O_2}\right)\frac{RT}{V}$$

$$= (10+4)\frac{(0.082 \times 473)}{20} = 27.15 atm$$

Final pressure $= \left(n_{H_2} + n_{O_2}\right)\frac{RT}{V}$

$$= \frac{(2+8)(0.082) \times 473}{20} = 19.30 atm$$

54. 00003.09

Sol. The difference in energy (ΔE) will be equal to the energy of the photon emitted.

The first line in Lyman series corresponds to the transition $2 \rightarrow 1$

$$\Delta E = 2.18\times10^{-18}(2)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)\text{J per atom}$$

$$= 6.54\times10^{-18}\text{ J}$$

Using the Einstein's photoelectric equation

$$KE = E_i - W_0 = \frac{1}{2}m_e v_e^2 \qquad [E_1 = \text{Incident energy}]$$

$$\Rightarrow v_e = \sqrt{\frac{2(E_i - W_0)}{m_e}}$$

$$= \sqrt{\frac{2\left(6.54\times10^{-18} - 2.18\times10^{-18}\right)}{9.1\times10^{-31}}} = 3.09\times10^6\text{ m s}^{-1}$$

55. 00002.67

56. 00004.47

57. 00038.85

Sol. According to Kohlrausch's law

$$\Lambda^{\infty}_{eq.CH_3COONa} = 91.6 = \lambda^{\infty}_{CH_3COO^-} = \lambda^{\infty}_{Na^+}$$

$$\Lambda^{\infty}_{eq.HCl} = 425.0 = \lambda^{\infty}_{H^+} = \lambda^{\infty}_{Cl^-}$$

$$\Lambda^{\infty}_{eq.NaCl} = 128.1 = \lambda^{\infty}_{Na^+} = \lambda^{\infty}_{Cl^-}$$

Adding eqs. (i) and (ii) and then, subtracting eq. (iii)

$$\lambda^{\infty}_{CH_3COO^-} + \lambda^{\infty}_{H^+} = \Lambda^{\infty}_{CH_3COONa} + \Lambda^{\infty}_{HCl} - \Lambda^{\infty}_{NaCl}$$

$$= 91.6 + 425.0 - 128.1$$

Or $\Lambda^{\infty}_{eq.CH_3COOH} = 388.5\text{ S cm}^2\text{ eq}^{-1}$

Also, $\alpha = \dfrac{\Lambda_v}{\Lambda^{\infty}}$

$$0.001 = \frac{\Lambda_v}{388.5}$$

$$\Lambda_v = 38.85\text{ S cm}^2\text{ eq}^{-1}$$

58. 00036.79

Sol. Given, $\Delta H = 0.12\times10^3\ cal$;

$$E_{a(b)} = 0.02\times10^3\ cal \quad (E_a\ can\ never\ be\ negative)$$

$$\Delta H = E_{a(f)} - E_{a(b)}$$

$$E_{a(f)} = 0.12\times10^3 + 0.02\times10^3\ cal$$

$$= 0.14\times10^3\ cal$$

% of molecules crossing over the barrier

$$= 100\times e^{-E_{a(f)}/RT}$$

$$= 100\times e^{-140/(2\times70)} = 100\times e^{-1}$$

$$= 100\times0.8062 = 80.62\% = \frac{100}{2.718} = 36.79$$

59. 00001.87

Sol. Given, $K_4Fe(CN)_6 = \frac{1}{10}N$

$K_4Fe(CN)_6 = \frac{1}{10}\times\frac{1}{4}M \quad (N = M \times valence\ factor)$

For

$K_4Fe(CN)_6, n/V = \frac{1}{40} mol\ litre^{-1}$

$= \frac{1}{40}\times 10^3 mol\ m^3 = 25\ mol\ m^3$

$\because \pi_N \times V = nST \Rightarrow \pi_N = \frac{n}{V}ST$

$\because \pi_N = 25\times 8.314\times 300 = 6.236\times 10^4\ Nm^{-2}$

Where, α is degree of dissociation

Given, $\alpha = 0.5$

Now, $\pi_{exp}/\pi_N = (1+4\alpha)$

$= 6.236\times 10^4 (1+4\times 0.5)$

$= 1.87$

60. 00000.72

Sol.

	H_2	+	I_2	$\rightleftharpoons$	2HI
volume at $t=0$	60		42		0
Volume at equilibrium	$(60-x)$		$(42-x)$		$2x$

The volume of gas $\propto$ mole for a gas at same P and T

$\because K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2x)^2}{(60-x)(42-x)} \quad ....(i)$

For reactions $\Delta n = 0,$ one can use mole in place of concentration (mol litre^{-1}) since formula will not involve volume terms.

Given, $2x = 28 \quad \therefore x = 14$

By eqs. (i) and (ii)

$K_c = \frac{28\times 28}{(60-14)(42-14)} = \frac{28}{46} \quad .....(iii)$

Now for dissociation of HI

	2HI	+	H_2	$\rightleftharpoons$	I_2
mole at $t=0$	1		0		0
Mole at eq.	$(1-\alpha)$		$(\alpha/2)$		$(\alpha/2)$

Where α is degree of dissociation

$\therefore \quad K_{c_1} = \frac{(\alpha/2)\times(\alpha/2)}{(1-\alpha)^2} = \frac{\alpha^2}{4(1-\alpha)^2}$

$\therefore \quad \frac{46}{28} = \frac{\alpha^2}{4(1-\alpha)^2}\left(K_{c_1} = \frac{1}{K_c} = \frac{46}{28}\right)$

$\alpha = 0.72$

Mathematics

PART – III

SECTION – A

61. C

Sol. p_i must be divisible by 2 and 3 both.
Hence $p_1 = 6$, $p_2 = 2$, $p_3 = 3$.

62. C

Sol. Let us suppose A be origin.
$\text{ar}(\Delta ABC)^2 + \text{ar}(\Delta ACD)^2 + \text{ar}(\Delta ABD)^2 = \text{ar}(\Delta BCD)^2$.
Hence the result follows.

63. C

Sol. $T_r = 2^{r-1}$ as $a_1, a_2, a_3, \ldots, a_{r-1}$ are have exactly 2 possible values.
For $1 \le i \le r-1$ and $a_r = 1$ then

$$\therefore \lim_{r\to\infty} \frac{S_r}{T_r} = \lim_{r\to\infty}\left[\frac{1}{18}\left(1-\frac{1}{10^{r-1}}\right)+\frac{1}{10^r}\right] = \frac{1}{18}.$$

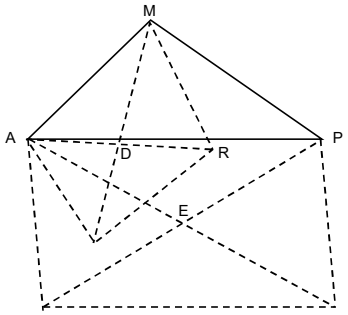
64. B

Sol. $$\frac{z_A - z_E}{z_P - z_E} = i$$

$$z_E = \frac{z_A - iz_P}{1-i}$$

$$\arg\left(\frac{z_P - z_M}{z_E - z_D}\right) = \arg\left(\frac{(z_P - z_M)(1-i)}{z_A - iz_P - z_A + iz_M}\right)$$

$= \arg(1 + i) = \frac{\pi}{4}$.



65. C

Sol. We know that $1 + 3 + 5 + \ldots + (2k + 1) = (k + 1)^2$.

66. D

Sol. RHD: $f'(k) = \lim_{h\to 0} \frac{f(k+h) - f(k)}{h}$

$$= \lim_{h\to 0} \frac{[2k+2h]\sin 3\pi(k+h) - [2k]\sin 3k\pi}{h}$$

$$= \lim_{h\to 0} \frac{[2k]\sin(3\pi k + 3\pi h)}{h} = 2k(-1)^k 3\pi = 6k\pi(-1)^k$$

$\Rightarrow \lambda = 6$.

67. A

Sol. Circumcircle of ΔABC passes through focus $\left(\frac{1}{4k+4}, 0\right)$ of the given parabola.

$\therefore$ Putting y = 0, in equation of circumcircle, we get either $\frac{1}{4k+4} = 2$ or $\frac{1}{4k+4} = 3$.

68. B

Sol. Number of terms = 50, so m = 50
f(x) will be minimum from x = 625 to x = 678
So, n = 54

69. D

Sol. We have $BC = I_B$

$\Rightarrow tr(A) + tr\left(\frac{A}{2}\right) + tr\left(\frac{A}{2^2}\right) + ... + tr\left(\frac{A}{2^n}\right) = \frac{tr(A)}{1-\frac{1}{2}}$

$= 2tr(A) = 6.$

70. D

Sol. Probability that they get correct solution is $\frac{1}{8} \times \frac{1}{12} = \frac{1}{96}$

Probability that they get wrong solution is 77 / 96 .

If E_1 be the event of both getting correct solution and E_2 be the event that both gets wrong solution.

Let E be event of both obtaining the same answer, then

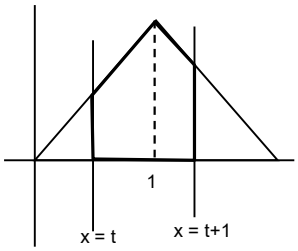
$$P\left(\frac{E_1}{E}\right) = \frac{P(E_1)P\left(\frac{E}{E_1}\right)}{P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)P\left(\frac{E}{E_2}\right)} = \frac{\frac{1}{96} \times 1}{\frac{1}{96} \times 1 + \frac{77}{96} \times \frac{1}{1001}} = \frac{13}{14}.$$

71. B

Sol. $ar(M \cap N) = -t^2 + t + \frac{1}{2} \ \forall \ t \in [0, 1]$

$ar(M \cap N) = -\left(t - \frac{1}{2}\right)^2 + \frac{3}{4}$

$\Rightarrow \min(ar(M \cap N)) = \frac{3}{4}$ sq. units.



72. A

Sol. $$\frac{(e^x - \sin x)(\cos x - 3)(\ln x + 2)}{\left(x^4 - x + \frac{1}{2}\right)(2x-7)^{30}} \le 0$$

$x^4 - x + \frac{1}{2} = \left(x^4 - x^2 + \frac{1}{4}\right) + \left(x^2 - x + \frac{1}{4}\right)$ = positive

Using wavy curve $x \in \left[\frac{1}{e^2}, \frac{7}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$.

73. B

Sol. The abscissae are: a_1, ma_1, m^2a_1 ..... which are in G.P.

Now, $A.H = \frac{a_1\left(\frac{m^n - 1}{m-1}\right)}{\frac{1}{a_1}\left(\frac{1-\left(\frac{1}{m}\right)^n}{1-\left(\frac{1}{m}\right)}\right)} = a_1^2 . m^{n-1} = G^2$

74. D

Sol. Let $f(x)=\dfrac{\sqrt{x^2+1}+x-1}{\sqrt{x^2+1}+x+1}$

$\Rightarrow f(x)=\dfrac{x^2+1-(x-1)^2}{\left(\sqrt{x^2+1}+x+1\right)\left(\sqrt{x^2+1}-x+1\right)}$

$f(x)=\dfrac{x}{\sqrt{x^2+1}+1}$

$\Rightarrow \int_1^2 \dfrac{xdx}{\sqrt{x^2+1}+1}.$

75. A

Sol. $I=\int_\alpha^\beta \dfrac{e^{f(a(\alpha-x))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx$ …(1)

$I=\int_\alpha^\beta \dfrac{e^{f(a(x-\beta))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx$ …(2)

From (1) and (2), we get

$\Rightarrow I=\dfrac{1}{2}(\beta-\alpha)$

76. B

77. B

Sol. Clearly $m>\dfrac{1}{4}$ and I = 0

78. C

Sol. R is not reflexive as (2, 2) $\notin$ R, (3, 3) $\notin$ R
Also, R is not transitive as (2, 1) $\in$ R and (1, 2) $\in$ R but (2, 2) $\notin$ R.
However R is symmetric.

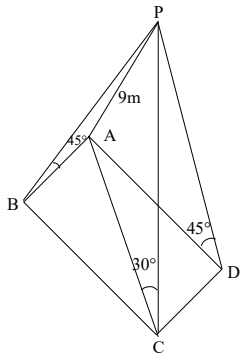
79. B

Sol.

p	q	$\sim p \vee q$	$(\sim p \wedge \sim q)$	$(\sim p \wedge q) \wedge (\sim p \wedge \sim q)$
T	T	T	F	F
T	F	F	F	F
F	T	T	F	F
F	F	T	T	T

80. A

Sol. Let AP be the lamp post of 9 m standing at corner A of the rectangular field ABCD



In $\Delta(S)\, BAP \,\&\, CAP$ we have $\tan 45° = \dfrac{PA}{BA}$ & $\tan 30° = \dfrac{PA}{AC}$

$\Rightarrow\ BA = 9m \,\&\, AC = 9\sqrt{3}m \Rightarrow BC = \sqrt{AC^2 - AB^2} = 9\sqrt{2}m$

Hence area of the field is $AB \times BC = 9 \times 9\sqrt{2}m^2 = 81\sqrt{2}m^2$

SECTION – B

81. 00120.00

Sol. $m = \lfloor 4 \rfloor 5$

$n = 4 \lfloor 4 \rfloor 5$

$\Rightarrow m + n = \left(\lfloor 5 \right)^2$

82. 00002.60

Sol. $\dfrac{x^8 - 1}{x - 1} = 0$ where $x \neq 1 \Rightarrow$ roots are $e^{i\pi/4}, e^{i2\pi/4}, e^{i3\pi/4}, ..., e^{i7\pi/4}$

$\Rightarrow$ Required are $= \left(\sqrt{2}\right)^2 + \dfrac{(4-1)\left(\sqrt{2}-1\right)}{2} = \dfrac{3\sqrt{2}+1}{2}\,\text{sq. units}$

Hence $a = 3, b = 2, c = 1, d = 2$

$\Rightarrow \dfrac{9+4}{5} = \dfrac{13}{5} = 2.6$

83. 00003.00

Sol. We have $\lim\limits_{x \to \infty} \dfrac{\int_0^x e^{m(t)}dt}{\dfrac{e^{m(x)}}{n(x)}} \dfrac{\infty}{\infty}$ form

84. 00000.60

Sol. Clearly $I = \sum\limits_{n-0}^{\infty} \int_0^1 \left(x^{3n} - x^{3n+1}\right)dx = \dfrac{\pi}{3\sqrt{3}} = \dfrac{3.1415}{5.196}$

85. 00000.15

Sol. Required area $= \int_1^3 (x-1)^2 (x-2)^2 (x-3)^2\, dx = \dfrac{16}{105}$

86. 00010.25

Sol. $\bar{z}=\frac{\sum x}{n_1}, \bar{y}=\frac{\sum y}{n_2}$ as there are 18 observations

Combined mean $=\frac{\left(a_1\times 10\right)+\left(a_2\times 8\right)}{18}$ where a_1, a_2 denotes are means but

$a_1=15 \Rightarrow a_2=\frac{45}{4}$

87. 00024.83

Sol. $\vec{a}+\vec{b}=6\vec{p}-\vec{q} \quad \left|\vec{a}+\vec{b}\right|=15$

Similarly $\left|\vec{a}-\vec{b}\right|=\left|4\vec{p}+5\vec{q}\right|=\sqrt{617}$

88. 00001.68

Sol. Let $\mathrm{L}=\lim_{n\to\infty}\left(\frac{{}^{3n}\mathrm{C}_{2n}}{{}^{2n}\mathrm{C}_{n}}\right)=\lim_{n\to\infty}\left[\left(1+\frac{n}{n+1}\right)\left(1+\frac{n}{n+2}\right)---\left(1+\frac{n}{(n+n)}\right)\right]^{1/n}$

$\ln \mathrm{L}=\int_0^1 \ln(x+2)\,dx-\int_0^1 \ln(x+1)\,dx$

$\Rightarrow \mathrm{L}=\frac{27}{16}$

89. 00016.00

Sol. Required area is area bounded by $\left[|x|\right]+\left[|y|\right]=3$

90. 00002.15

Sol. Apparently $\frac{\sin 2x}{\sin 2y}=\frac{3}{2} \& \frac{\cos 2x}{\cos 2y}=\frac{-19}{29}$

Hence $\frac{\sin 2x}{\sin 2y}-\frac{\cos 2x}{\cos 2y}=\frac{3}{2}+\frac{19}{29}=\frac{87+38}{58}=2.15$